

Leave management system

Create a app Employee in your project

In settings.py

```
INSTALLED_APPS = [  
    'django.contrib.admin',  
    'django.contrib.auth',  
    'django.contrib.contenttypes',  
    'django.contrib.sessions',  
    'django.contrib.messages',  
    'django.contrib.staticfiles',  
    'Employee',  
]
```

Models.py

```
from django.db import models  
from django.utils import timezone  
  
class Employee ( models.Model ) :  
    emp_id = models.AutoField( primary_key=True )  
    emp_name = models.CharField( max_length=50)  
    emp_phone = models.CharField( max_length=20)  
    emp_email = models.EmailField()
```

```
def __str__( self ) :  
    return f'{self.emp_id} - {self.emp_name}'
```

```
class Leave_Master ( models.Model ) :  
  
    leave_id = models.AutoField( primary_key=True)  
    leave_name = models.CharField( max_length=20)  
  
    def __str__( self ) :  
        return f'{self.leave_id} - {self.leave_name}'
```

```
class Leave_Employee_Assign ( models.Model ) :  
  
    sl_no = models.AutoField( primary_key=True)  
    emp_id = models.ForeignKey(Employee, on_delete=models.CASCADE)  
    leave_id = models.ForeignKey(Leave_Master, on_delete=models.CASCADE)  
    no_of_assigned_leave = models.IntegerField()  
    no_of_aval_leave = models.IntegerField()  
  
    def __str__( self ) :  
        return f'{self.emp_id} - {self.leave_id}'
```

```
class Leave_Apply ( models.Model ) :  
  
    sl_no = models.AutoField( primary_key=True )  
    emp_id = models.ForeignKey( Employee, on_delete=models.CASCADE)  
    date_of_leave = models.DateField( default=timezone.now)  
    leave_emp_id = models.ForeignKey( Leave_Employee_Assign, on_delete=models.CASCADE)  
    no_of_apply_leave = models.IntegerField()
```

```

def __str__ ( self ) :
    return f'{self.emp_id} - {self.leave_emp_id}'

def save(self, *args, **kwargs):
    if self.no_of_apply_leave <= self.leave_emp_id.no_of_aval_leave :
        self . leave_emp_id . no_of_aval_leave -= self.no_of_apply_leave
        self . leave_emp_id . save ()
        return super(Leave_Apply, self).save(*args, **kwargs)
    else :
        raise ValueError(f'No of aval leave is {self.leave_emp_id.no_of_aval_leave}')

```

admin.py

```

from django.contrib import admin
# Register your models here.
from . models import *
admin . site . register ( Employee )
admin . site . register ( Leave_Master )
admin . site . register ( Leave_Employee_Assign )
admin . site . register ( Leave_Apply )

```

views.py

```

from django.shortcuts import render
from . models import Employee , Leave_Apply , Leave_Master, Leave_Employee_Assign

```

Create your views here.

def leave_apply (request) :

empls = Employee . objects . all ()

leaves = Leave_Employee_Assign . objects . all ()

msg = None

if request.method == 'POST' :

emp_id = request . POST ['emp_id']

leave_type = request.POST ['leave_type']

no_of_apply_leave = int (request.POST ['no_of_apply_leave'])

try :

emp_id = Employee . objects . get (emp_id = emp_id)

leave_id = Leave_Master . objects . get (leave_name = leave_type)

leave_emp_id = Leave_Employee_Assign . objects . get (emp_id = emp_id , leave_id=leave_id)

lobj = Leave_Apply (emp_id = emp_id, leave_emp_id= leave_emp_id,
no_of_apply_leave=no_of_apply_leave)

lobj . save ()

msg = 'Leave Applied Success'

except Exception as obj :

msg = f'Leave Applied Failed - {obj}'

return render (request , 'leave_apply.html' , context={'empls':empls, 'msg' : msg, 'leaves' :
leaves})

leave_apply.html

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Leave Application</title>

  <style>

    body {

      font-family: Arial, sans-serif;

      background-color: #f4f4f9;

      margin: 0;

      padding: 20px;

      display: flex;

      justify-content: center;

      align-items: center;

      min-height: 100vh;

    }

    .container {

      background-color: #fff;

      padding: 20px;

      border-radius: 8px;

      box-shadow: 0 0 15px rgba(0, 0, 0, 0.1);

      max-width: 600px;
```

```
width: 100%;  
}
```

```
h2 {  
    text-align: center;  
    color: #333;  
    margin-bottom: 20px;  
}
```

```
form p {  
    margin-bottom: 15px;  
}
```

```
label {  
    display: block;  
    margin-bottom: 5px;  
    color: #555;  
    font-weight: bold;  
}
```

```
select, input[type="number"], input[type="submit"] {  
    width: 100%;  
    padding: 10px;  
    margin-top: 5px;  
    border: 1px solid #ddd;
```

```
border-radius: 4px;  
box-sizing: border-box;  
}
```

```
input[type="submit"] {  
    background-color: #4CAF50;  
    color: white;  
    border: none;  
    cursor: pointer;  
    font-size: 16px;  
    transition: background-color 0.3s ease;  
}
```

```
input[type="submit"]:hover {  
    background-color: #45a049;  
}
```

```
table {  
    width: 100%;  
    border-collapse: collapse;  
    margin-top: 20px;  
}
```

```
th, td {  
    text-align: left;
```

```
padding: 12px;  
border-bottom: 1px solid #ddd;  
}
```

```
th {  
    background-color: #f2f2f2;  
}
```

```
tr:hover {  
    background-color: #f5f5f5;  
}
```

```
.message {  
    margin-top: 20px;  
    padding: 10px;  
    background-color: #e7f3fe;  
    border-left: 6px solid #2196F3;  
    color: #31708f;  
    font-weight: bold;  
}
```

```
@media (max-width: 600px) {  
    .container {  
        padding: 15px;  
    }  
}
```

```

    }
</style>
</head>
<body>
    <div class="container">
        <h2>Leave Application</h2>
        <form method="post">
            <input type="hidden" name="csrfmiddlewaretoken"
value="ChJfEPgeWgduSkXrL2UjyvABnr5ei3pitIeShcEiqKVbK8i5zVO2SVEL5O7RKsLi">
            <p>
                <label for="id_emp_id">Emp Name:</label>
                <select name="emp_id" required="" id="emp_id">
                    <option value="" selected="">-----</option>
                    {% for obj in empls %}
                        <option value="{{ obj.emp_id }}">{{ obj.emp_name }}</option>
                    {% endfor %}
                </select>
            </p>
            <p>
                <label for="id_leave_emp_id">Leave Type:</label>
                <select name="leave_type" required="" id="leave_type">
                    <option value="" selected="">-----</option>
                    <option value="Earned Leave">Earned Leave</option>
                    <option value="Comp Off Leave">Comp Off Leave</option>
                    <option value="Medical Leave">Medical Leave</option>

```

```
<option value="Paid Leave">Paid Leave</option>

</select>

</p>

<p>

    <label for="id_no_of_apply_leave">No of apply leave:</label>

    <input type="number" name="no_of_apply_leave" required=""
id="no_of_apply_leave">

</p>

<input type="submit" value="Submit">

</form>

{% if msg %}

    <div class="message">

        {{ msg }}

    </div>

{% endif %}

<table>

<thead>

    <tr>

        <th>Emp Name</th>

        <th>Leave Name</th>

        <th>Leave Assigned</th>

        <th>Leave Remaining</th>

    </tr>
```

```

</thead>

<tbody>

    {% for obj in leaves %}

        <tr>

            <td>{{ obj.emp_id.emp_name }}</td>

            <td>{{ obj.leave_id.leave_name }}</td>

            <td>{{ obj.no_of_assigned_leave }}</td>

            <td>{{ obj.no_of_aval_leave }}</td>

        </tr>

    {% endfor %}

</tbody>

</table>

</div>

</body>

</html>

```

Urls.py

```

from django.contrib import admin

from django.urls import path

from Employee . views import leave_apply

urlpatterns = [

    path('admin/', admin.site.urls),

```

```
path('leave/', leave_apply ),  
]
```

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